

REPORT 1

**Project Title: Service Swap (Subscription-Based
Service Exchange Platform)**

INFO 3135: Web Development Using PHP & MySQL

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Due date:04/02/2026

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1. Introduction

Service Swap operates as an online platform which allows users to exchange their abilities and services without making any financial transactions. The platform implements a system which uses credits to handle transactions as users earn credit through their service work and use credits to request assistance from other users. Service Swap enables users to exchange services through its platform which uses digital security measures to protect user profiles and service information and credit management and messaging and review and dispute resolution functions. The platform monetizes through optional subscription plans rather than charging users per transaction. The document presents a comprehensive analysis of the Service Swap project requirements which includes the description of the problem, identification of the target users, and documentation of both functional and non-functional requirements, and use case diagrams constructed with UML.

2. Original Objective

The original objective of the Service Swap project, as defined during project proposal and inception, is: To design and implement a scalable web application that allows users to exchange services using a structured credit or time-based system, which eliminates the need for direct monetary payments while maintaining trust and accountability and usability. The system is intended to: The system is intended to: The system is intended to: The system is intended to: The service access costs will be decreased by the system. The system will develop community skill-sharing programs. The system will create secure informal trading systems. The system will create secure informal trading systems. The system will create secure informal trading systems. The system will use subscription-based revenue models to enable all future expansion.

2. Target Users

The Service Swap platform is designed for the following user groups:

Primary Users

Freelancers use the platform to exchange their professional services which include design work, development services, writing capabilities, and marketing expertise. Students provide tutoring services and technical assistance to others while receiving mentorship and academic support in exchange for their services. Professionals work as accountants, consultants, coaches, and advisors. Local Community Members use the platform to exchange home services and creative services and personal services.

2 Administrative Users

System Administrators handle user management tasks while they also resolve disputes and moderate content and control subscriptions.

4. Functional Requirements

1 User Account Management

- Users need to register for an account and log into the system using secure methods.
- Users need the ability to update their profile information which includes their skills and bio details and their current availability status.
- Users should have access to view the profiles of other users.

4.2 Service Management

- Users need the ability to create service listings which they can modify and remove from the system.
- The system needs to organize services into categories which users can easily find.
- Every service needs to establish a credit value or time value which users must understand.

4.3 Service Exchange (Swap System)

- Users should have the ability to request services from other users.
- Service providers have the right to either accept or decline service requests.
- The system must secure credits in escrow until a request receives approval.
- Users need the ability to confirm that they have completed the service.
- The system will transfer credits only after users confirm that the service has been completed.

4.4. Communication

Users must use in-app messaging to handle their communication requirements. The users need complete process visibility to monitor their swap operation from beginning to end.

4.5 Trust and Safety

Users need to have the ability to create ratings and reviews of their experiences. Users need to have the ability to report conflicts between parties.

The administrators need tools that allow them to resolve and manage disputes between different parties.

4.6 Subscription Management

Users should be able to choose premium subscription plans.

The platform restricts user visibility while showing active users according to their current subscription status.

5. Non-Functional Requirements

- The system needs to enable numerous users to work together at the same time.
- The system needs to deliver web pages to users at speeds which meet their acceptable time requirements.

5.2 Security

- Passwords must be encrypted
- User input must be validated to prevent SQL injection and XSS
- Role-based access control must be enforced

5.3 Usability

- The system needs an interface which people can use without help through its design which enables smooth navigation.
- The system needs to provide accessible features which non-technical users can operate without difficulty.

5.4 Scalability

The database design needs to accommodate increasing user numbers and additional service requirements. The system architecture needs to provide developers with the ability to create new features.

5.5 Reliability

- All credit transactions need to maintain reliable records which can be verified through audit processes.
- All swap transactions from the past must maintain complete record-keeping for future reference.

6. UML Documentation Summary

.1 Use Case Diagram

The system's main user functions through its member role which serves both users and service providers. The system includes administrative functions which only an Admin user can perform.

Core Member Use Cases:

- Register / Login
- Manage Profile
- Browse / Search Services
- View Service Details

- Create / Edit Service Listing
- Request Service
- Accept / Decline Request
- In-App Messaging
- Track Swap Status
- Confirm Completion
- Escrow & Transfer Credits
- Ratings & Reviews
- Report / Dispute

Administrative Use Cases:

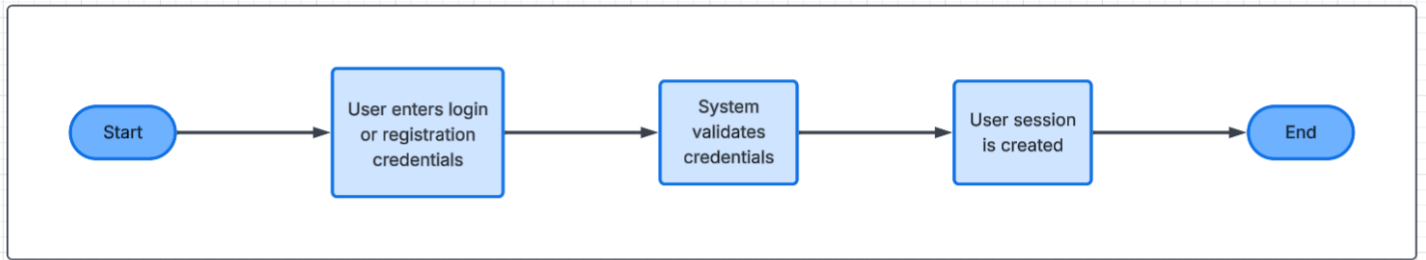
- Manage Users
- Moderate Listings & Categories
- Resolve Disputes
- Manage Subscriptions

6.2 Textual Use Case Descriptions

Use Case 1: Register / Login

- Primary Actor: Member
- Description: The system provides users with the ability to create new accounts or log into their existing accounts.
- Precondition: The user must be logged out of the system.
- Postcondition: The user receives authentication and proceeds to the dashboard.
- Main Flow:
 - o The user provides either login information or registration details
 - o The system performs validation of the provided credentials
 - o The system establishes a new user session for the user

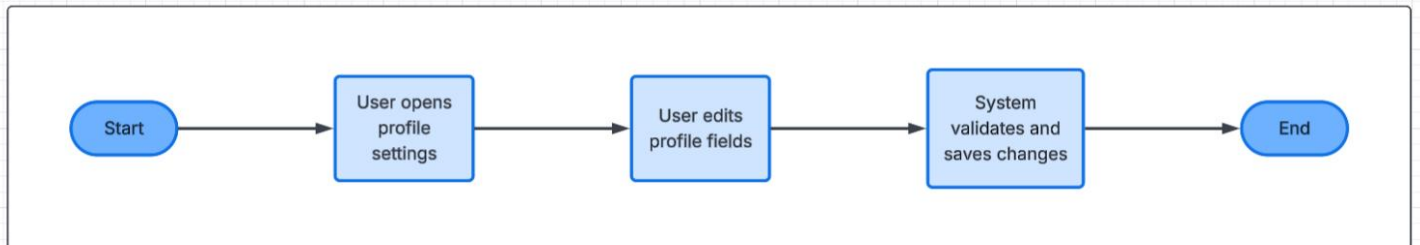
Register/Login Use Case Diagram



Use Case 2: Manage Profile

- The main character in the story is the Member role.
- The system allows users to modify their profile information through their skills and bio details and their available time slots.
- The user must be authenticated to access the system.
- The system will store newly updated profile information as permanent data.
- The Main Flow sequence starts when the user accesses profile settings.
- The user proceeds to make changes in their profile information.
- The system checks the changes for accuracy and proceeds to store them.

Manage Profile Use Case Diagram

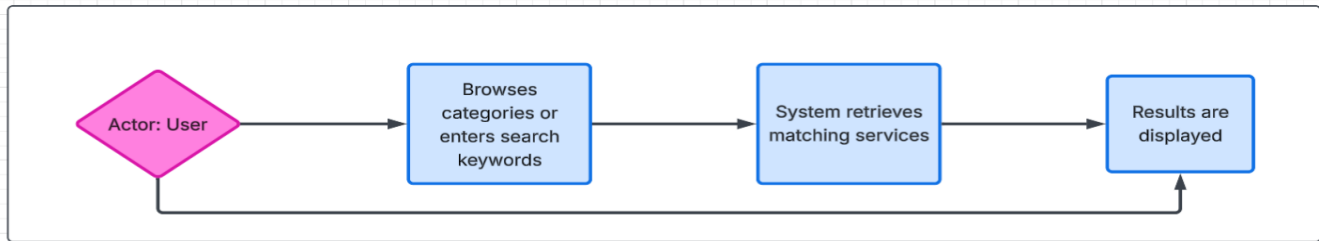


Use Case 3: Browse / Search Services

Primary Actor: Member

- Description The system enables users to find available services through its category and search filter functions.
- Precondition The user must be authenticated to access the system.
- Postcondition The system shows all services that match the search criteria.
- Main Flow User browses categories or enters search keywords The system retrieves all services that match the user's query The system shows the results to the user.

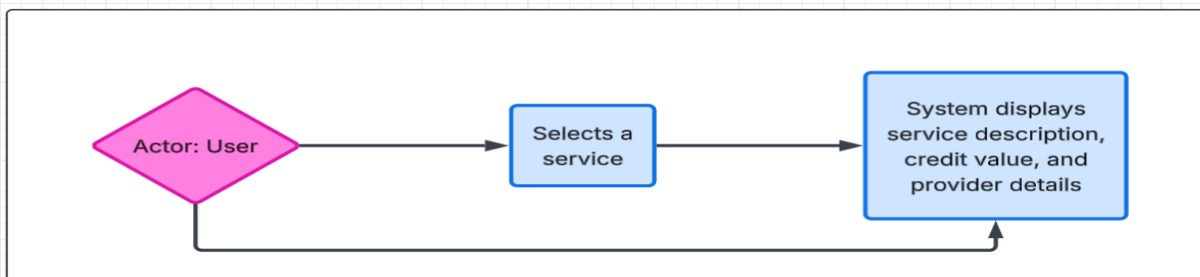
Browse/Search Services Use Case Diagram



Use Case 4: View Service Details

- Primary Actor: Member
- Description: The system shows complete information about the selected service listing.
- Precondition: Service exists in the system.
- Postcondition: Service details are shown.
- Main Flow:
 - o User selects a service
 - o System displays service description, credit value, and provider details

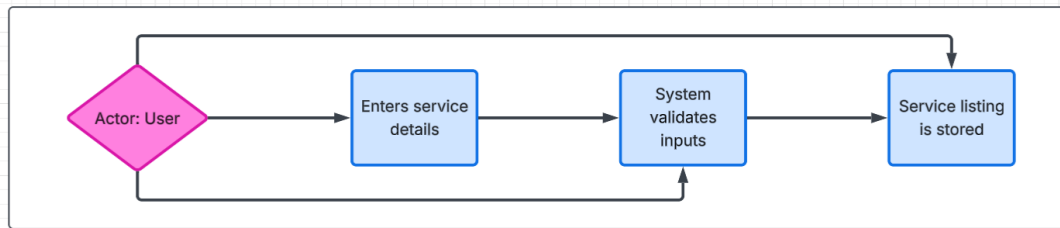
View Service Details



Use Case 5: Create / Edit Service Listing

- Primary Actor: Member (Service Provider)
- Description: Users can create and modify their service offerings.
- Precondition: User must be authenticated through their account.
- Postcondition: The service listing undergoes either a new save or an existing update.
- Main Flow: User enters service details
 - o System validates inputs
 - o Service listing is stored.

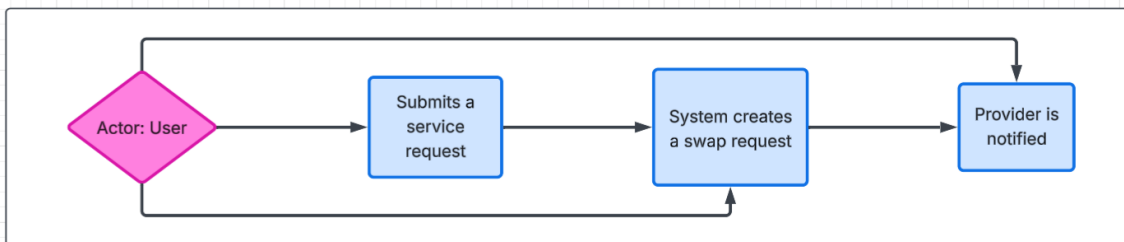
"Create/Edit Service Listing"



Use Case 6: Request Service

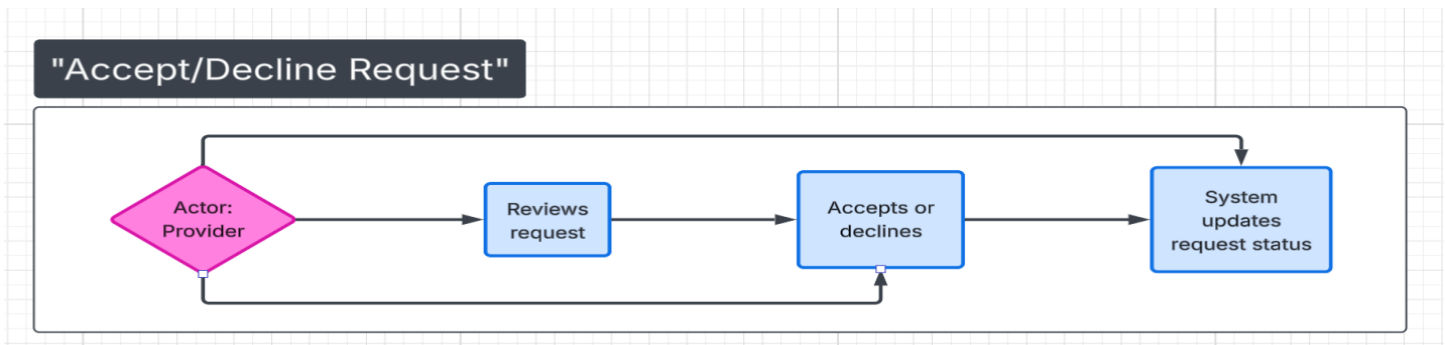
- Member serves as the primary actor of this system.
- The system allows users to request services which other users will provide.
- The user needs to have sufficient credits for advancement.
- The system generates a service request which displays the Requested status.
- Main Flow: User submits a service request System creates a swap request Provider is notified.

Request Service



Use Case 7: Accept / Decline Request

- The main actor of the system is a member who acts as a service provider.
- The system enables service providers to reply to incoming service requests which they receive.
- The system requires an existing service request before it can operate.
- The system updates the request status as its result.
- The main process starts when the provider examines the request.
- The provider must choose between accepting the request or declining it.
- The system changes the status of the request after the provider plans.



Use Case 8: In-App Messaging

- The member functions as the main actor of the system.
- The system enables users to talk with each other during their service exchange.
- The system requires an existing swap request as its initial condition.
- The system stores all messages while it sends them to their intended recipients.
- The main process starts when a user sends a message to the system.
- The system proceeds to send the message to its intended recipient after the user has sent it.

Use Case 9: Track Swap Status

- Main actor of the system: Member
- Platform Description: The platform enables users to track the progress of service exchanges.
- User Swap Requirement: The user needs to participate in a swap to use the system.
- The system displays: The active swap status to users.
- Main Flow:
 - o the user checks their ongoing swaps
 - o the system shows the present service status to users.

Use Case 10: Confirm Completion

- The main actor of the system functions as a member.
- The system works to authenticate successful completion of a service.
- The system requires service delivery to complete its precondition.
- The system changes swap status to Completed after the process ends.
- The process flows when the user verifies that he finished his work.
- The system controller changes the swap status after the user confirms his work.

Use Case 11: Escrow & Transfer Credits

- **Primary Actor:** System
- **Description:** Handles to secure credit transfer between users.
- **Precondition:** Swap marked as completed.
- **Postcondition:** Credits are transferred to provider.
- **Main Flow:**
 - System releases escrowed credits
 - Provider wallet is credited

Use Case 12: Ratings & Reviews

- **Primary Actor:** Member
- **Description:** Allows users to rate and review complete exchanges.
- **Precondition:** Swap is completed.
- **Postcondition:** Review is stored and visible.
- **Main Flow:**
 - User submits rating and comment
 - System saves review

Use Case 13: Report / Dispute

- **Primary Actor:** Member
- **Description:** Allows users to report issues with a service exchange.
- **Precondition:** Swap exists.
- **Postcondition:** Dispute is logged for admin review.
- **Main Flow:**
 - User submits dispute
 - System records dispute

Use Case 14: Administrative Management

- **Primary Actor:** Admin
- **Description:** Allows administrators to manage users, content, disputes, and subscriptions.
- **Precondition:** Admin is authenticated.
- **Postcondition:** Administrative action is applied.
- **Main Flow:**
 - Admin selects management tasks
- System applies changes